

Dot Product

I. Definition

Vector Operation 2: Dot Product. Let $\mathbf{a} = [a_1, a_2, a_3]$ and $\mathbf{b} = [b_1, b_2, b_3]$. Then

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

- An easier way to compute the dot product of 2 known vectors is to use:

$$\mathbf{a} \cdot \mathbf{b} = [a_1, a_2, a_3] \cdot [b_1, b_2, b_3] = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Geometrical Interpretation:

Dot Product

$$\mathbf{a} \cdot \mathbf{b} = (\text{component of } \mathbf{a} \text{ along } \mathbf{b})(\text{length of } \mathbf{b})$$

II. Additional Concepts

Notes:

- $\mathbf{a}, \mathbf{b} \in \mathbf{R}^3 \implies \mathbf{a} \cdot \mathbf{b} \in \mathbf{R}$ (*scalar*)

Properties: Let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbf{R}^3$ be non-zero.

- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$ or equivalently $|\mathbf{a}| = \sqrt{\mathbf{a} \cdot \mathbf{a}}$
- Commutative: $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
- Distributive: $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{c}$

Thm: Let $\mathbf{a}$ and $\mathbf{b}$ be non-zero vectors. Then

$$\mathbf{a} \cdot \mathbf{b} = 0 \quad \text{iff} \quad \mathbf{a} \text{ and } \mathbf{b} \text{ are perpendicular (orthogonal)}$$

Important Inequalities:

- Triangle Inequality: $|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$
- Cauchy-Schwarz Inequality: $|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}||\mathbf{b}|$

III. Application

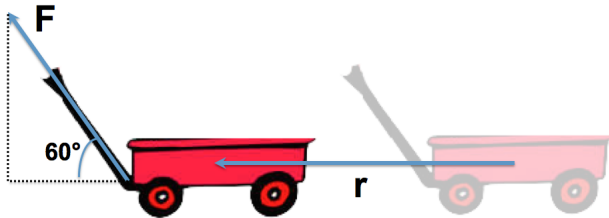
Work: In the most basic form (1-Dimension), work on an object is given by:

$$\text{Work} = \underset{\text{(on an object)}}{FORCE} \cdot \underset{\text{(object moved)}}{DISTANCE}$$

However, this assumes:

- Constant Force
- Force is acting in the same direction as the movement of the object.

Now let $\mathbf{F}$ be a force pulling a wagon and $\mathbf{r}$ the position of the wagon.



In this case, the force is not acting in the same direction as the movement of the wagon.

To find the work, we must decompose the force to find out how much force is contributing to the movement.

Since $|\mathbf{F}| \cos \theta$ is the quantity of the force in the direction of movement and $|\mathbf{r}|$ is the total distance traveled then the work to pull the wagon is

$$Work = (Force)(Distance) = (|\mathbf{F}| \cos \theta)(|\mathbf{r}|) = |\mathbf{F}||\mathbf{r}| \cos \theta = \mathbf{F} \cdot \mathbf{r}$$

Therefore, for any constant force, $\mathbf{F}$, the general expression for work done is a **dot product**

$$Work = \mathbf{F} \cdot \mathbf{r}$$

Ex: If the wagon is being pulled by a constant force of $[-5, 10] N$ and the displacement of the wagon is $[-4, 0] m$, then what is the work being done on the wagon?

$$Work = [-5, 10] \cdot [-4, 0] = (-5)(-4) + (10)(0) = 20 \text{ N}\cdot\text{m}$$